

Machine Learning vs. Deep Learning

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Assignment: 2.3 Machine Learning vs. Deep Learning

Introduction

Machine learning and deep learning are related branches of artificial intelligence, but they differ in how they process data and learn patterns. Traditional machine learning usually depends on humans to select meaningful features before an algorithm is trained. Deep learning uses multilayer neural networks that can identify important features directly from raw data. Traditional machine learning is often appropriate for structured datasets with clearly defined variables, while deep learning is better suited for complex data such as images, video, audio, and natural language. The lesson also explains that deep learning generally requires more data and computational resources than traditional machine learning.

This report examines house-price prediction as an example of traditional machine learning and facial image recognition as an example of deep learning. It explains a real-world application of each approach, why the selected approach is suitable, and why the alternative approach would be less appropriate.

Traditional Machine Learning Example: House-Price Prediction

Description and Real-World Application

House-price prediction is a traditional machine learning problem that can be addressed using a regression algorithm, such as linear regression. A model can be trained using historical property data and features such as square footage, number of bedrooms, number of bathrooms, property age, neighborhood, school quality, lot size, and recent sale prices. The model learns how these variables relate to the final selling price and then estimates the value of another property.

A real-world application is an automated property-valuation system used by real-estate companies, mortgage lenders, property investors, and online housing platforms. For example, a lender could use a machine learning model to produce an initial estimate of a property's market value before ordering a formal appraisal. A real-estate website could also use the model to provide homeowners with estimated property values based on recent market data.

Why Traditional Machine Learning Is Suitable

Traditional machine learning is suitable for house-price prediction because housing data is usually structured and organized into clearly defined variables. Features such as square footage, bedroom count, ZIP code, property age, and previous sale price can be represented in rows and

columns. These variables have understandable relationships with the target value, which is the estimated sale price.

Regression algorithms are effective because they can measure how strongly each feature influences the prediction. For example, the model may determine that additional square footage generally increases a property's value, while an older property may require an adjustment based on its condition. The prediction can also be explained by showing which variables contributed most to the result.

Traditional machine learning offers several practical advantages for this problem. It can perform well with a moderately sized dataset, requires less computing power than deep learning, and is typically faster to train. It is also more interpretable. Interpretability is important in real estate and lending because professionals may need to explain how a valuation was produced.

The course lesson identifies house-price prediction as a machine learning example and explains that features such as bedrooms, square footage, and neighborhood quality can be provided to a linear regression model to predict a house's price.

Why Deep Learning Is Less Suitable

Deep learning could technically be used for house-price prediction, but it would not usually be the most suitable first choice when the available data consists mainly of structured property features. A deep neural network may contain many layers and parameters, requiring a larger dataset, more computational resources, and longer training time.

For a relatively straightforward regression problem, this additional complexity may not produce a meaningful improvement over a traditional algorithm. A deep learning model may also be more difficult to interpret. If a mortgage lender or homeowner asks why a property received a certain valuation, explaining the decision of a complex neural network may be harder than explaining the coefficients or feature importance of a traditional model.

Deep learning would become more appropriate if the valuation system also analyzed complex unstructured information, such as interior photographs, aerial images, floor plans, or written property descriptions. However, for a model based primarily on structured numerical and categorical features, traditional machine learning is more efficient, transparent, and practical.

Deep Learning Example: Facial Image Recognition

Description and Real-World Application

Facial image recognition is a deep learning problem commonly addressed using a convolutional neural network, or CNN. A CNN processes raw image pixels and learns visual patterns through multiple layers. Early layers may identify simple features such as edges and lines, while deeper layers may recognize textures, shapes, facial structures, and relationships among facial features.

A real-world application is facial authentication on smartphones and secure devices. When a user attempts to unlock a phone, the system captures an image of the user's face and compares the learned facial representation with an enrolled reference. Similar technology can be used for building access, identity verification, photo organization, and fraud prevention.

The lesson identifies image recognition as a deep learning example and explains that CNNs learn directly from raw pixel data. They progressively identify patterns such as edges, textures, and complex facial features without requiring humans to define every visual feature manually.

Why Deep Learning Is Suitable

Deep learning is suitable for facial image recognition because images contain highly complex and unstructured data. A single image may contain thousands or millions of pixel values. Facial appearance can also change because of lighting, camera angle, distance, facial expression, hairstyle, glasses, aging, or partial obstruction.

It would be extremely difficult for a human designer to manually define every feature needed to recognize a face under all these conditions. A CNN solves this problem by automatically learning a hierarchy of visual features from large collections of labeled images. Instead of relying only on predefined measurements, the network discovers which patterns are most useful for distinguishing one face from another.

The multilayer structure of a CNN is especially valuable. Initial layers detect basic visual elements, while later layers combine these elements into increasingly complex representations. This allows the model to learn that a face is not defined by a single feature, but by the spatial relationships among many features.

Deep learning models can also improve as they are trained with more varied data. Images captured under different lighting conditions, angles, expressions, and backgrounds can help the system generalize to real-world environments. This makes deep learning more appropriate than a simpler model for a task involving raw visual data and complex patterns.

Why Traditional Machine Learning Is Less Suitable

Traditional machine learning would require significant manual feature engineering for facial recognition. A developer might attempt to define features such as the distance between the eyes, the shape of the nose, the width of the mouth, or the outline of the face. These measurements could then be supplied to an algorithm such as a support vector machine.

However, manually designed features may fail when a person's face is viewed from a different angle, partially covered, poorly lit, or captured at a different resolution. Human-selected features may also overlook subtle patterns that help distinguish individuals.

A traditional machine learning system could work in a controlled environment with standardized images and a limited number of users. However, it would be less effective for a large-scale real-world facial recognition system. The process of creating and maintaining manual features would

be difficult, and the model would likely be less adaptable to the many variations found in image data.

Deep learning avoids much of this limitation by learning relevant features directly from the images. Therefore, although traditional machine learning may be simpler and less computationally expensive, it is not as suitable for the complexity and variability involved in facial image recognition.

Comparison of the Two Approaches

The main difference between the two examples is the nature of the data. House-price prediction primarily uses structured data with variables that can be clearly identified in advance. Facial recognition uses unstructured image data in which the important visual patterns are difficult to define manually.

Traditional machine learning is appropriate for house-price prediction because the features are understandable, the dataset may be moderate in size, and interpretability is valuable. Deep learning would add unnecessary complexity unless the system also processed images or other unstructured information.

Deep learning is appropriate for facial recognition because neural networks can automatically extract complex visual features from raw pixels. Traditional machine learning would require extensive manual feature engineering and would struggle to account for the wide range of variations found in real-world images.

These examples demonstrate that deep learning is not automatically better than traditional machine learning. The appropriate approach depends on the problem, the type and quantity of available data, computational requirements, interpretability needs, and the complexity of the patterns being learned.

Conclusion

House-price prediction and facial image recognition illustrate why different artificial intelligence problems require different modeling approaches. House-price prediction is well suited to traditional machine learning because it uses structured variables, can be solved efficiently with regression algorithms, and benefits from interpretable results. Deep learning is generally unnecessary for this type of problem unless complex unstructured data is added.

Facial image recognition is well suited to deep learning because images contain complex patterns that are difficult to represent through manually selected features. Convolutional neural networks can automatically learn visual characteristics from raw pixel data and adapt to differences in lighting, angle, expression, and appearance.

Selecting the correct approach requires more than choosing the most advanced technology. It requires matching the model to the characteristics of the problem. Traditional machine learning

provides simplicity, efficiency, and interpretability for structured prediction tasks, while deep learning provides the feature-learning capacity needed for complex and unstructured data.